

Linear Algebra

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Intro

The notes here are based off of the book "Linear Algebra Done Right".

1 Vector Spaces

Notation 1. $\mathbb{F}$ stands for either $\mathbb{R}$ or $\mathbb{C}$. That is, if a theorem holds for $\mathbb{F}$, it holds for both the real and complex numbers.

The elements of $\mathbb{F}$ are called 'scalars'. Lists of length n are ordered collections of n objects. Cannot be infinite, must have the same order to be classed as equal. Also often called an n -tuple.

Definition 1. $\mathbb{F}^n$ is the set of all lists of length n of elements of $\mathbb{F}$:

$$\mathbb{F}^n = \{(x_1, \dots, x_n) : x_k \in \mathbb{F} \text{ for } k = 1, \dots, n\}.$$

For $(x_1, \dots, x_n) \in \mathbb{F}^n$ and $k \in \{1, \dots, n\}$ we say that x_k is the k th coordinate of $(x_1, \dots, x_n)$.

Addition is commutative in $\mathbb{F}^n$. There exist additive inverses in $\mathbb{F}^n$. Scalar multiplication in $\mathbb{F}^n$ is such that we have $\lambda \in \mathbb{F}$ and $x \in \mathbb{F}^n$ to get $\lambda x \in \mathbb{F}^n$.

Definition 2. A vector space is a set V along with an addition on V and a scalar multiplication on V such that we have commutativity, associativity, additive identity, additive inverse, multiplicative identity, and distributive properties.

Example 1. An example of a vector space involves a set of functions:

- If S is a set, then $\mathbb{F}^S$ denotes the set of functions from S to $\mathbb{F}$.
- For $f, g \in \mathbb{F}^S$, the sum $f + g \in \mathbb{F}^S$ is the function defined by

$$(f + g)(x) = f(x) + g(x),$$

for all $x \in S$.

- For $\lambda \in \mathbb{F}$ and $f \in \mathbb{F}^S$, the product $\lambda f \in \mathbb{F}^S$ is the function defined by

$$(\lambda f)(x) = \lambda f(x),$$

for all $x \in S$.

For instance, if $\mathbb{F} = \mathbb{R}$ and $S = [0, 1]$ then $\mathbb{R}^{[0,1]}$ is the set of all real valued functions on the interval $[0, 1]$.

A vector space has a unique additive identity. Every element in a vector space has a unique additive inverse.